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Evaluation: Does conformance information derived from a process model improve the prediction of future deviations and compliance risks?

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Contents

1 Introduction

In the following we will evaluate two different experimental studies to compare and evaluate the usage of conformance information in the predictions of future deviations and compliance risks. Later, we will discuss the overall viability of the usage of conformance information in predictive process mining tasks.

2 Experimental Study 1: BPI Challenge 2012

1. A clear definition of the prediction problem and label construction: for example, defining a case as "deviant" if its final alignment cost exceeds a chosen threshold, if a particular rule is violated (e.g., an LTL constraint), or if conformance checking detects a specific type of non- compliant pattern. Motivations for the chosen definition must be documented. Lab Evaluation Report: deliverables and content

1 Definition of the prediction problem and label construction for conformance deviation targets

explaining both technical details and conceptual implications for organizations that wish to anticipate and mitigate non-compliant behavior.

2 Choosing dataset for the prediction problem

3 definition of the dedicated CLI walkthrough -¿ Maybe 2-3 cases which are based on prefix length

4 evaluation protocol: How will the data be processed and what splitting etc. is applied and why exactly that

5.1 Application of Baseline Model

5.2 Application of Conformance-aware model (which may augment the results) -¿ Clearly state the chosen parameters and why they were chosen for the specific usecase

6 Evaluate the target with the evaluation module's outputs.

7 COMPARISON: baseline/conformance target predictions -¿ Usage of visualizations and metrics

Discussion on computational aspects: Measure, reporting of runtime, feasability -¿ mention the conformance deviations -¿ mention that if prefix-generation with rising length makes the whole process taking much more computational power

3 Experimental study 2: Sepsis Cases

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4 viability discussion

8 General discussion: -¿ does it improve predictions using conformance features? -¿ Which indicators are useful -¿ threats to validity? ——¿ Mention alignment noise, multiple target labels for one case, definition of labels, missing labels for cases, bad data quality, highly individual data, -¿ limitations and ideas to future work